NEW CHAPTER

# Three Major Challenges When Facing Transformation

When an organization starts a major transformation, the biggest problems are usually not technical.
They are behavioral and organizational.

### The "Firefighting" Trap

Teams focus on fixing surface problems every day.

- Systems fail → Fix it quickly
- Complaints appear → Respond immediately
- Deadlines slip → Push harder
- But they never fix the root cause.
- Result: Energy is spent on crises, not on real transformation.

### Change Burnout

People have seen too many "new teams or new leaders".
They think:

- "We've heard this before."
- "Let's wait and see."
- "It will fail anyway."
- When employees stop believing, even good strategies fail.
- Real change requires trust and visible results — not slogans.

### False Urgency

Every goal feels urgent.

- Too many priorities
- Too many projects
- No clear focus
- Organizations confuse "being busy" with "making progress."
- Real urgency means focus, clarity, and direction.

# The Rise & Fall of NPfIT — And How GDS Was Born

The UK's National Programme for IT (NPfIT) launched in 2002 became one of the world's most infamous IT failures. Later, **GDS (Government Digital Service) was created largely to correct the mistakes NPfIT made** — a direct response born from institutional failure.

BACKGROUND

# What Was NPfIT?

A high-budget project (£12 billion+) led by the Labour government to fully digitize the National Health Service (NHS). Executed by NHS Connecting for Health (CfH, internal unit), the programme was officially terminated in 2011 — widely regarded as the UK's largest civilian IT failure.

### Core Goal

Build a single, centralized national electronic health record, e-referral system (Choose and Book), and network infrastructure.

### Execution

Led by NHS Connecting for Health (CfH), relying heavily on large outsourced contractors.

### Outcome

Terminated due to massive budget overruns, technical delays, and frontline clinician refusal to adopt the system.

# Where to Start?

## How Did GDS Rescue the Digital Transformation Effort

LESSONS LEARNED

# GDS vs. NPfIT: Old Thinking vs. New Thinking

When GDS launched in 2011, its 10 Design Principles were a direct antidote to every failure mode of NPfIT. In short, **GDS exists so that disasters like NPfIT never happen again** — shifting government IT from giant long-term contracts with big vendors **to internal design teams using agile** and open-source.

| | NPfIT Failure (Old) | | GDS Principle (New) |
|---|---|---|---|
| ❌ | Top-down: officials & big vendors decide features | ✅ | **#1 Start with user needs** — ask patients & doctors what they actually need |
| ❌ | Big Bang: build the entire national system at once | ✅ | **#5 Iterate. Then iterate again** — ship small, test, learn, repeat |
| ❌ | **Closed & bureaucratic:** contracts & architecture is not transparent | ✅ | **#10 Make things open**: it makes things better — **open code & standards** |
| ❌ | **Complex design:** systems so hard to use clinicians rebelled | ✅ | **#4 Do the hard work to make it simple** — absorb complexity so users don't have to |

**Key takeaway:** GDS (internal unit) transformed how the UK government procures technology — **from outsourcing everything to building in-house capability** with agile, user-centered, and open-source approaches.

# GDS's Ten Design Principles

01

Start with user needs

02

Do less

03

Design with data

04

Do the hard work to make it simple (for user)

05

Iterate, then iterate again

06

This is for everyone

07

Understand context

08

Build digital services, not websites

09

Be consistent, not uniform

10

Open makes things better

*these principles emerged from design practice.

# Why the Ten Design Principles Matters?
# –Proving GDS Was Serious About Change

### Balancing Act

Too radical and they would be considered as impractical ideals; too safe and they would be pulled back by the force of bigger, older institutions, and then nothing changes.

### Standard Practice

GDS's ten principles were standard practice for an organization growing up under the principles of the open internet, but they were not standard government behavior.

Initially, few officials paid attention to these principles. The first audience was those peering in from outside government. The principles sent a clear signal to digitally skilled talent: the new team—and the UK government—were serious.

# Start Small

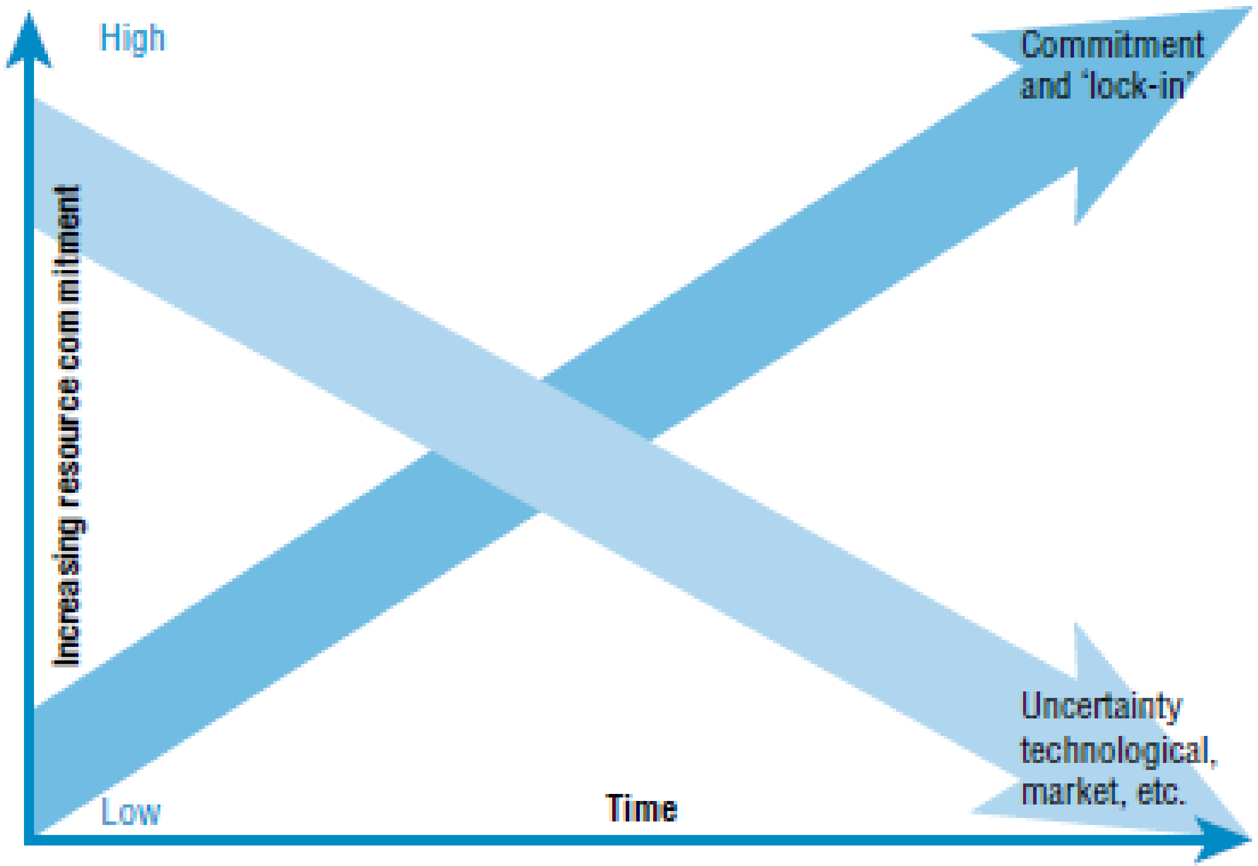
Once you've shown the world how you intend to operate, you'll be in a much better position to decide where to begin. The first challenge for any new digital team is to demonstrate to observers that it can deliver working outcomes on the web faster than the organization has ever managed before.

Your strategy should be delivery (Getting Things Done):  Producing working code that can actually be deployed must be the top priority, over writing elegant strategy documents, defining organizational structures, or arranging office space. All of these are important, but secondary to demonstrating something of value to real users outside the organization.

*** Delivery is the best form of marketing: in large transformations, the “bystanders” are budget holders and political leaders — earning their trust secures the political capital needed for survival.**

# Uncertainty vs. Resource Commitment

Economically, **innovation is viewed as a process of reducing uncertainty while increasing resource commitment**, here is the classic diagram (Tidd & Bessant, 2020).




- **Uncertainty (technical, market, etc.) decreases over time**
- **Resource commitment increases over time**
- **As commitment increases, it creates a "lock-in effect," making it harder to change direction**

* The Dark side of Lock-in effect:
The more you invest, the harder it becomes to walk away — even if you should.

# The True Meaning of Fast

In some organizations, fast might mean a year, or 18 months at most.

13

Weeks

Time to build a working alpha version of GOV.UK

11

Weeks

Time for the UK e-petitions service to go from zero to live

These were government projects, remember. This doesn't mean that these services were completely finished; all services are still being iteratively improved today. They all started small, simple, clearly designed, and user-tested.

Your instinct should be to make work public when it still feels a little uncomfortable. This can be difficult, especially for government workers. The work is not perfect, but improve continuously with user feedback — and most people prefer fast, imperfect progress over no change at all.

# Petitions & E-Petition Services

### What does "petition" mean?

A **petition** is a formal request made by a group of people to someone in authority.

It usually means:

Many people sign their names to show they want something to change.

### What is an e-petition Service?

An **e-petition service** is an online platform that allows citizens to:

- Create petitions
- Collect digital signatures
- Trigger official government responses when support reaches a certain threshold

It is a digital tool that lowers the barrier for public participation in policymaking.

# Four Questions for Choosing Your First Project

The first two are about impact, and the second two are about risk. Your first projects should deliver tangible benefits (visible, usable, measurable, communicable) to users while carrying little to no political risk.

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| **How many people will benefit, and by how much?**<br><br>The goal is to deliver small but visible improvements related to a large number of people quickly. For example, enabling citizens to renew their driver's license under 3 minutes on a mobile phone instead of visiting a government office. | **Does this solve a common problem?**<br><br>You will quickly discover that organizations solve the same problem multiple times in different places. Ideally, you want to create a service that effectively meets a specific user need and can then be easily adopted and adapted by other teams. Ex., centralized login service that all departments can reuse and adapt. | **What is the institutional complexity?**<br><br>The number of teams involved influences completion times. This grows when those teams are from different organizations or departments. The practical guideline for government projects is to add six months for every additional department. | **Is it greenfield or brownfield?**<br><br>Greenfield services are built to address new (or newly identified) user needs. Brownfield services come with varying degrees of existing expectations. If possible, start with a new small service (ex., e-petition), or one that is completely beyond repair<br>➔ Why are people reluctant to start with a brownfield project? |

# Sources of Good Ideas

If you're out of ideas, various people within the organization will be more than happy to offer some. But be careful about accepting these ideas without understanding their motivations; they might have nothing to do with the actual needs of the organization's users.

## Most Reliable Sources — Staff

The most reliable sources for new digital service ideas are often the operational staff on the front lines, closest to the public. They what is broken in the current process and have learned how to work around the problems just to get things done.

*** Workaround-** A temporary way to avoid a problem without actually fixing the root cause ➔ In the long term, it will become technical debt.

# Sources of Good Ideas

## Most Reliable Sources — Data

### Web Traffic Analysis

Pinpoint **low-traffic pages** among thousands to prioritize content improvements.

### Customer Service Data

Uncover gaps **— what users are searching for but failing to find online**.

### Unfiltered Complaints

Collect **raw user complaints** to prevent those with louder voices from gaining false priority.

Digital teams may need to be creative in obtaining this data **— colleagues in other departments can be reluctant to share, especially if they haven't addressed the issues themselves**. In the UK, **Citizens Advice** charity data has exposed numerous government operational problems that some officials chose not to investigate.

# Key Takeaways：Design Principles for Delivery

### Start **Small & Visible**

Begin with small, **greenfield**, simple, and high-visibility projects — it's both **better and safer**.

### **Avoid Political Fixes**

Don't get pulled into patching organizational politics. Stay focused on delivery outcomes.

### Ship Fast, Iterate Often

**Release a working service to users** — internal or external — **quickly,** then **continuously improve through iteration**.

### Listen to the **Front Line**

Harvest the best ideas from **operational staff closest to the public** — they see what users truly need.

CHAPTER

# The First Team: The Power of Diversity

**Once you’ve defined how the work should be done and what needs to be delivered, you need to find the right people to do it.**

# Government Is Full of Smart People

Government officials often get a bad reputation. This is deeply unfair. **Most** bureaucracies are filled with very smart, talented, and committed people. The problem is not the individuals — it's caused by the organizational characteristics. Together, their output often falls far below the sum of its individuals.

At the heart of most government agencies, you'll find a group known as policy generalists/ generalist official. These are cross-disciplinary individuals who:

- Write well and are comfortable with numbers
- Have strong backgrounds in economics and history
- Can tackle almost any problem and craft solutions that look elegant on paper or in economic models

Many large corporations rely on similar minds. The talent is real — but so is the risk of building an entire organization around one type of thinker.

THE DANGER OF SAMENESS

# When Everyone Sees the World the Same Way

Even though these generalists are often brilliant, organizations that rely too heavily on one type of analytical mind now face a real and growing problem.

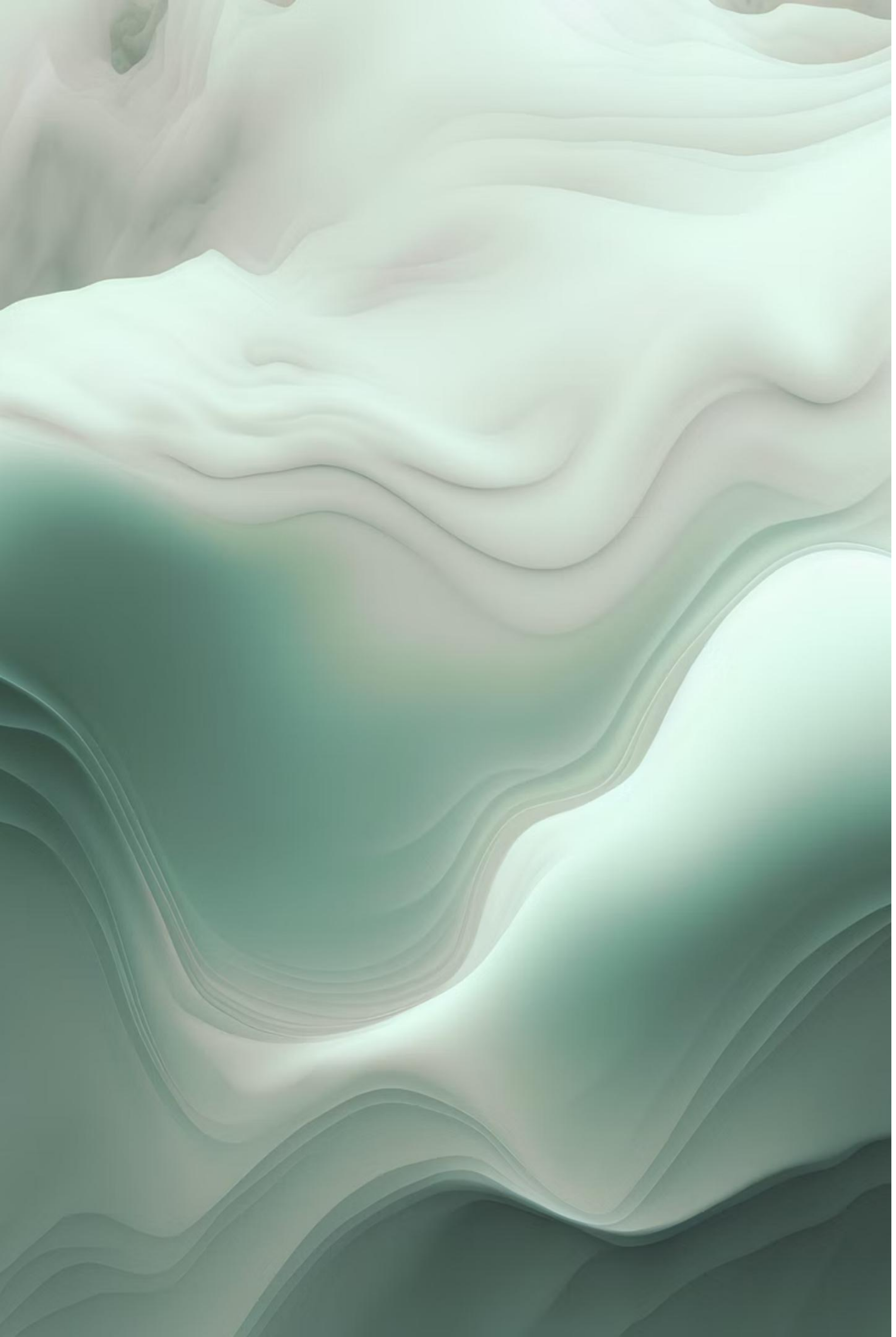
### Same Lens, Same Answers

When all leaders frame problems through the same perspective — and look to past generations of similar thinkers for validation — there is a strong tendency to apply old thinking to new problems.

### People who disagree get shut down

Those who push back find it a painful and potentially career-damaging choice. Over time, they either go quiet or leave entirely.

As the internet age advances, an organizational model built around one type of person becomes increasingly dangerous.



THE IMBALANCE

# What Happens When You Value Only Some Skills

Organizations that have prized certain qualities over others for many years **find it especially hard to build a viable digital transformation. That preference encourages one set of skills to thrive — while devaluing other strengths.**

### Extremely Strong

**Policy & Economics** — In the UK government, these two disciplines have long been the most valued skills — so **the system works best on problems that can be solved through these lenses.**

### Undervalued

**Operations, Design & Technology** — For challenges requiring these different perspectives, the picture is far less encouraging.

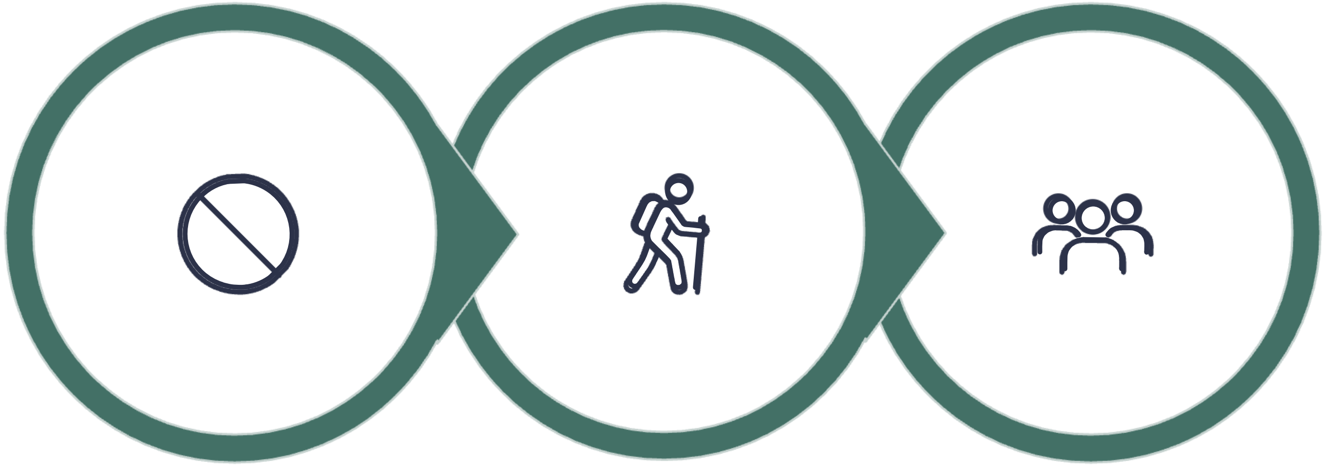
This is also a critical issue in other industries or sectors.

*** Viable: ‘usable’, ‘doable’, and ‘valuable’**

THE REAL MEANING OF TRANSFORMATION

# Digital Transformation Is About Balance, Not Replacement

This imbalance is not unique to the UK or to government. It happens whenever organizations make it hard for people with different professional strengths to rise to positions of influence.



Blocked from leadership

Talent leaves

Perspectives lost

The result? Organizations lose the very viewpoints they need most.

> Digital transformation does not mean replacing old skills with new ones. It means balancing old and new skills — and making them work together.

CASE

# Organ Donation Review: The Power of Digital Transformation

In 2012, the UK government began experimenting with new methods to increase organ donor numbers. The Behavioural Insights Team ("Nudge Unit") believed the key to success was modifying the driver's license application form (paper-based) to include a call to action encouraging donation. However, logistical difficulties in updating the paper-based process slowed progress, and frustration mounted between policy and operational departments.

# Discovery of the Golden Page

## The Dilemma of Paper Forms

Modifying driving license application form **was not only operationally difficult but also proved to be a bad idea** in practice. Most applicants were teenage drivers or their parents, who were not interested in sentences about death at the top of the form. Test results showed that the number of donations actually decreased.

## Digital Solution

After the launch of GOV.UK, the team discovered the potential of the "Golden Page" **– the last page of a transaction (ex., after renew license or passport)**. This was the moment people were most likely to extend their actions after completing a purchase, offering an excellent opportunity for public good.

# Advantages of Scaled Testing

- Creating multiple standardized versions of the end page is easier and cheaper than printing thousands of different forms.
- Tracking completion rates is also more direct.
- And most importantly, it makes large-scale testing of micro-policy interventions immediately and possibly.

* Compare above performance with modifying driving license form (paper-based): 20 million vs. ten thousands.

# Advantages of Scaled Testing

* Compare the effects of different solution versions.

### Social Norms

"People like you are donating organs" - leveraging group behavior to influence individual decisions

### Visual Elements

Some versions use images to enhance emotional connection and visual appeal

### The Give-and-Return Rule

"You might need these organs one day" - this strategy ultimately prevailed

# Astonishing Results

**400K**

New Donors

400,000 new organ donors in one year

**8**

Test Versions

Eight different versions tested on different landing pages

**Weeks**

Completion Time

The entire project completed in a few weeks

# Why The Power of Diversity Matter: Skills Reshaping in the Digital Age

The digital age has created some truly new roles, or at least redefined existing ones.

The problem is that governments and large organizations outsource skills they don't value to consultants or vendors. Over time, internal experts leave, and the organization becomes dependent on outside vendors.

## The Outsourcing Trap

1

### Loss of Expertise

Undervalued experts leave the large organization, often moving to suppliers.

2

### Knowledge Gap

The organization becomes an "uninformed" buyer of professional services.

3

### Inertial Dependence

Officials rely on supplier arrangements that seemed work in the past.

4

### Disastrous Contracts

Thus, 10-year, nine-figure IT disaster contracts are born
* In NPfIT case it's **£12 billion+.**

# Why The Power of Diversity Matter

Generalist officials and corporate strategists, on their own, genuinely try to solve problems, but often lack the complete toolkit needed to do so. This is where the value of diverse teams lies.

When you build products or services with national or global impact, your team needs to look like the audience you are trying to reach:

- Ask different questions
- Challenge hidden assumptions
- Prioritize accessibility (ex., for elders)
- Design beyond your own experience

# Breaking Down Departmental Walls
# : Why Agile Teams Must Ignore Organizational Walls

Organizational design is typically the product of a powerful mix of inertia, power dynamics, office politics, and leadership behavior. Institutions operating at government scale tend to slice themselves into as many silos as possible.

### Departmental Walls

Government-sized institutions organize themselves into as many silos as possible, with activities broken down into progressively smaller policy buckets.

### Skill Layering

Org may put different experts —IT, HR, economics, science, policy, etc.— side by side, but not truly integrate their work.

### Need for Flexibility

Agile teams need to dynamically reallocate people based on priorities — rather than waiting for departmental approval to address problems.

**There are only two golden rules for building the right team:**

- If everyone on the team is good at different things, you're probably on the right track
- If the team can solve different types of problems over time, that's even better

➔ simply introducing new skills is not enough — the real trick is placing these experts into agile, cross-functional teams

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A **silo** in an organization:

- Barriers between departments
  - Teams don’t talk to each other
  - Information is not shared
  - Each department protects its own territory

# Minimum Viable Team

CORE ROLES

Your first digital product will define the trajectory and work content of the digital agency. **Your first digital team will define the work culture and way of doing things.** Here are a few key positions you need to get started.

# Product Manager

The Head of the Team

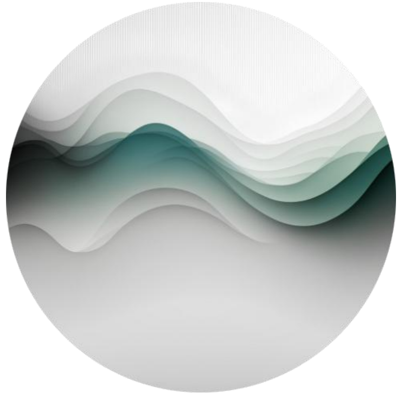
The Product Manager is the first person on the team and **the public face of the project**.

- They **define and articulate the product vision** being built, **explaining it to the team and the broader organization**.
- As **the voice of the product's users**, they must also understand the **operating environment**.
- They need to be able to judge **when to compromise and when to fight for the user**.
- The Product Manager **ultimately decides what to build, when to build it, and chooses the next hypothesis for the team to test**.

# Delivery Manager

### Translator of the Vision

The Delivery Manager is the **complementary partner to the Product Manager**. While the Product Manager articulates to the world what the team is doing, **the Delivery Manager translates Product Manager's vision into concrete tasks for the team.**



### Problem Solver

They are the **team's fixer, removing obstacles that prevent the team from getting work done –** from finding the right person to fill team vacancies, to mediating any clashing personalities.

# Lead Developer & Designer

## Lead Developer

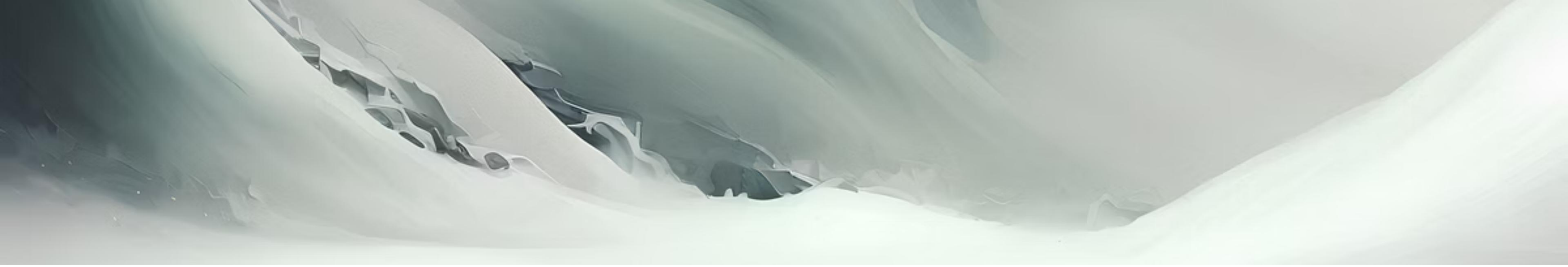
**Lead developers are the engine room of the team**. They focus on user needs and how services are consumed to build software. Their job is to write, tune, maintain, and support code, and to resolve technical issues.

The ideal developer for **an agile team cares more about building things that work for users** than about the programming language used. They should be knowledgeable about the latest programming languages, but not overly concerned with the cutting-edge code of the moment.

## Designer

Designers ensure that **the service presents a clear, consistent experience to users.** Depending on the project and maturity, different design skills will come to the fore.

**For small, early-stage teams, an interaction designer who can do some front-end coding is valuable.** As the service involves more dependencies within the organization, a service designer with a big-picture view becomes increasingly valuable.



# User Researcher

User researchers are responsible for providing the team with reliable, regular feedback from users and ensuring that the team understands how that feedback translates into changes for what they are designing or operating.

They will find and recruit research participants, conduct research sessions (which everyone on the team attends), and interpret the results. User research is very different from typical government consultation activities. A skilled researcher knows that the true value of user feedback is not in what they say, but in the gaps between the words.

> When GDS began developing an early version of GOV.UK, we were rightly criticized for not having a user researcher on the team initially. This was a big mistake—user researchers are an essential member of any good digital service team.

# Content Designer

## A New Role in the Text Industry

We take it for granted that government is a text industry. Thousands of officials do little but write. However, writing for officials is a very different skill from writing for the web.

There are 5 million functionally illiterate adults in the UK, reading below an 11-year-old's level. **Content designers are responsible for making the written content on services meet the needs and expectations of users.** For mainstream services, this means writing for everyone.

Government tends to value clever over clear content. Most users don't need clever. Like user research, content design wasn't given enough emphasis at the beginning of GDS until a brilliant woman defined the discipline and gave it the place it deserved within service teams.

# Finding Your Team

01

## The Power of Networks

**Start with your networks to find similar-minded people.** In the UK, many of the first government hires came from groups who had been interested in taking on this task for years.

02

## Engage with Communities

**Attend tech meetups and hackathons.** These rooms are full of people who give up their free time and scarce skills to solve the very problems you want to solve.

03

## Internal Talent

Don't forget the **brilliant people already within your organization.** Their job titles might not match what you're looking for, but the best people will adapt and become exemplars of these professions.

# Working Mode: Agile

Agile has as many definitions as there are people who have heard of it. **What's important is:**

- A team focused on user needs
- Delivering small, incremental improvements, failing fast, iterating, constantly planning for the future
- Thinking about how to improve the team's way of working

Agile work creates a steady and predictable way of working for the team:

- Short daily stand-up meetings at the start of each day
- One or two-week **'sprints'** with clear priorities and goals
- Regular reflections on what's working and what needs improvement

* A **sprint** is a short work period (usually 1–2 weeks) where a team **focuses on completing a specific set** of tasks (usually a small, usable result).

❏ **Agile rejects** the model, loosely generalized as 'waterfall,' as a **one-size-fits-all answer.**

# Working Mode: Openness and Flat Structure

## Open: Being transparency

From the outset, your first team must **strongly favor working openly. Publishing code** and design patterns on **GitHub** or similar platforms is crucial. **Publish regular updates** instead of sending internal memos.

## Flat Structure

Agile teams perform best as small and self-organizing units. Teams are groups of **peers**, and the vast majority of strategic decisions are **discussed collectively.** The product manager's role in these discussions is that of **a facilitator and referee, not a dictator**.

## Co-location

Product teams must work full-time in the same space. **Physical co-location offers huge advantages**, especially in the early stages of service delivery when the learning curve is steep and new information is abundant **(saving communication cost).**

*** Peers: People at the same level who respect each other's expertise**

# Key Takeaways

### Cross-functional Teams

Cross-functional teams, combining traditional business roles with digital skills, are crucial for successful digital agencies.

### Teams as Delivery Units

The unit of delivery is the team, not the individual. All team practices should strongly lean towards open, agile, flat, and co-located ways of working.

### Finding the Right People

Find the right people; don't expect them to find you. Seek out trusted networks and expert communities.

### Open, Flat, Co-location

Teams are groups of peers, and the vast majority of strategic decisions are discussed collectively.